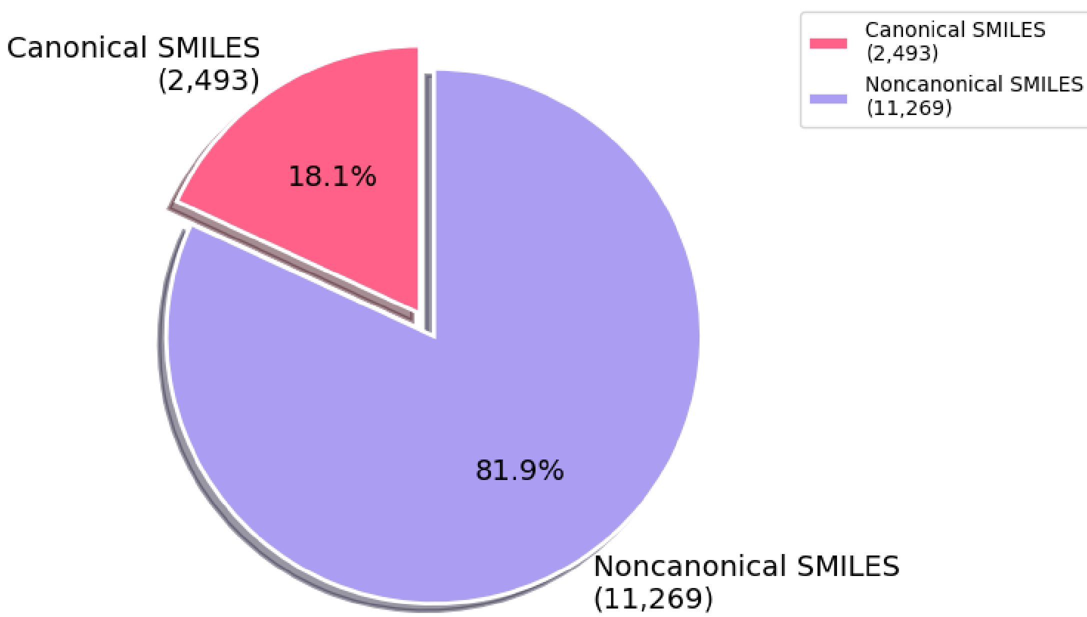
SMILES-Exploratory-Data-Analysis

March 17, 2026

1 SMILES Exploratory Data Analysis Report

1.1 Canonical SMILES Check

Intention: Determine the distribution of SMILES in the dataset that are canonical and non-canonical.

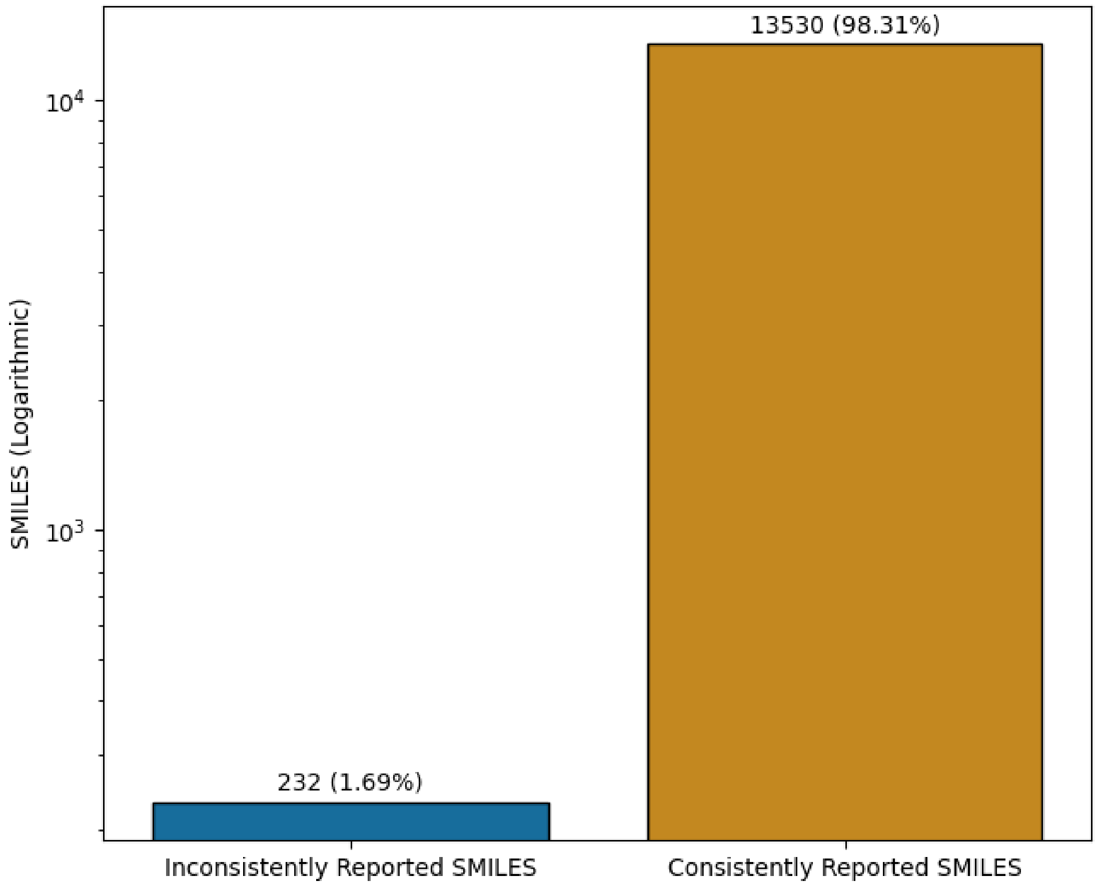


Findings: 2,493 SMILES were canonical (18.1% of dataset) and 11,269 SMILES were non-canonical (81.9 % of dataset)

Interpretation: Because non-canonical SMILES exist in the dataset, convert them to canonical SMILES. This is done to ensure that all SMILES are represented uniquely, leading to SMILES syntax consistency.

1.2 Inconsistently Reported SMILES Check

Intention: Determine the distribution of SMILES that were inconsistently reported and consistently reported. Inconsistently reported SMILES are classified as duplicate SMILES in the dataset with conflicting BBB permeability results.

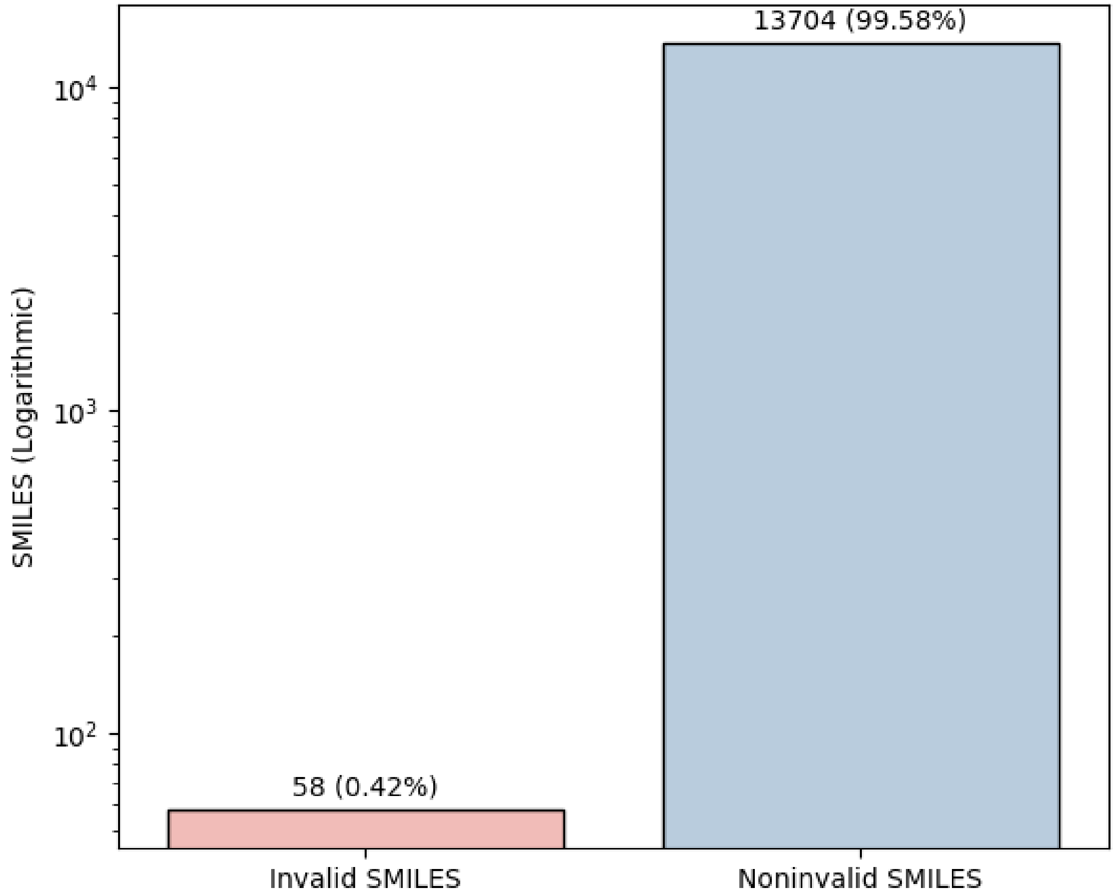


Findings: 232 SMILES were inconsistently reported (1.69 % of dataset) and 13,530 SMILES were consistently reported (98.31% of dataset)

Interpretation: Because it is impossible to determine which BBB permeability label is accurate for inconsistently reported SMILES, remove all instances of the SMILES. This is done for better model understanding and dataset accuracy.

1.3 Invalid SMILES Check

Intentions: Determine the distribution of invalid SMILES and valid SMILES in the dataset.

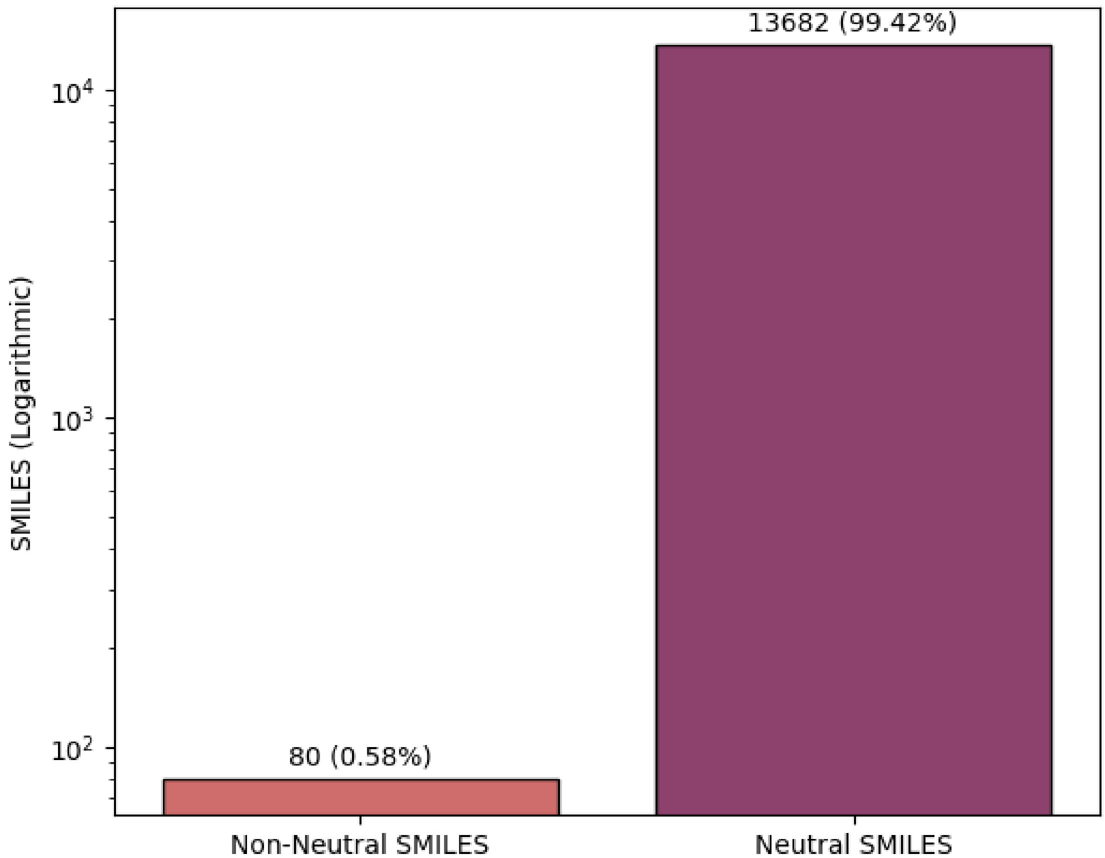


Findings: 58 SMILES were invalid (0.42% of dataset) and 13,704 SMILES were valid (99.58% of dataset)

Interpretation: Invalid SMILES often root from inaccurate syntax, and therefore should be removed due to uninterpretability. Removing invalid SMILES will remove corrupted SMILES data, increasing model understanding of SMILES structure.

1.4 Non-Neutral SMILES Check

Intentions: Determine the distribution of SMILES that are non-neutral and neutral in the dataset

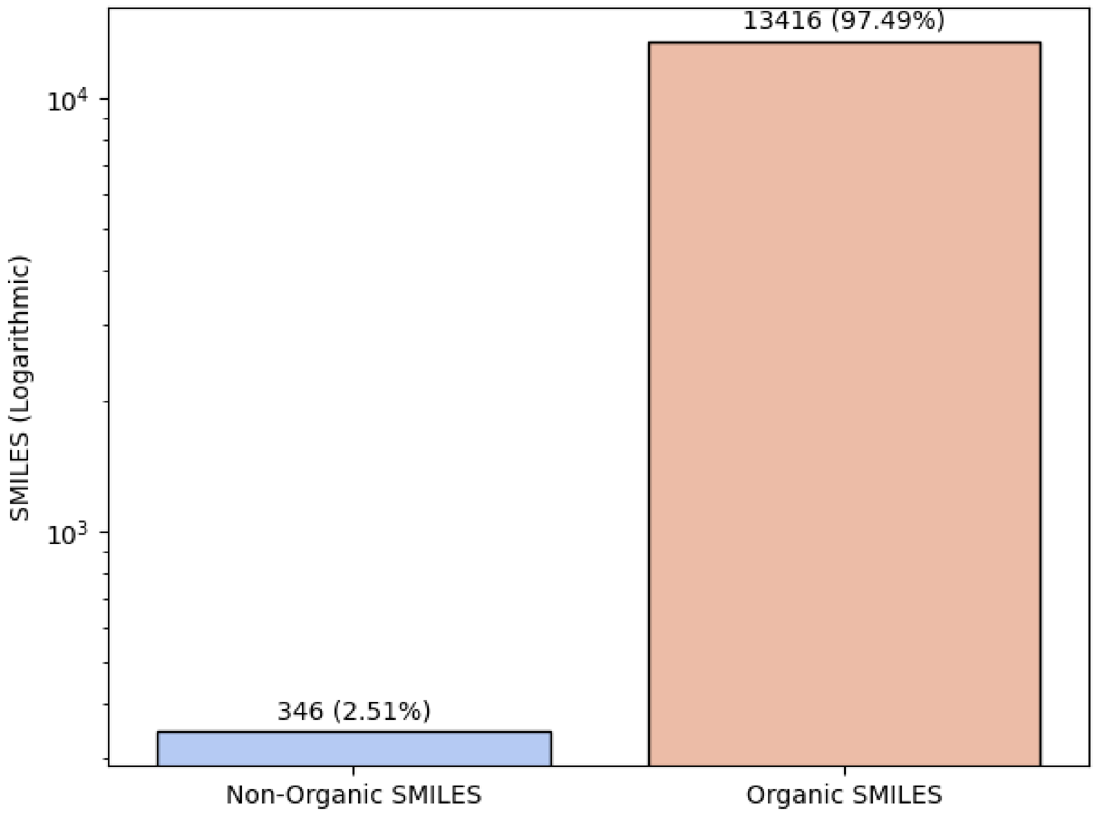


Findings: 80 SMILES were non-neutral (0.58% of dataset) and 13,682 SMILES were neutral (99.42% of dataset)

Interpretation: The SMILES with charges should be neutralized for standardization with the rest of the dataset. Keeping the charge will introduce noise to the dataset, resulting in poor model accuracy.

1.5 Non-Organic SMILES Check

Intentions: Determine the distribution of SMILES that are non-organic and organic in the dataset. Non-organic compounds are classified as compounds with atoms that have an atomic number greater than 20.

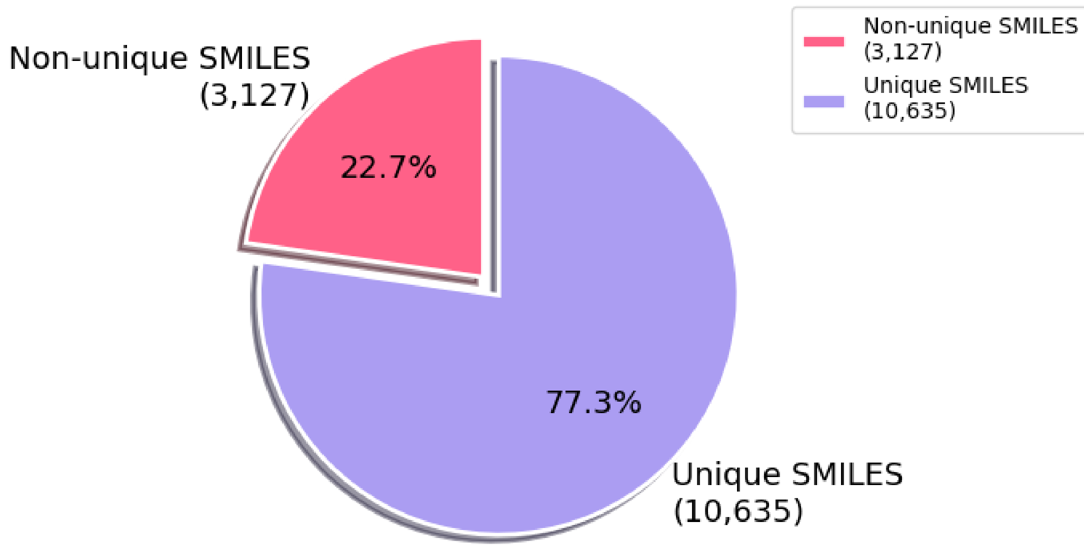


Findings: 346 SMILES were non-organic (2.51% of dataset) and 13,416 SMILES were organic (97.49% of dataset)

Interpretation: The non-organic SMILES in this dataset should be removed because the model's primary focus is to predict the permeability of small organic compounds. Therefore, the non-organic SMILES will introduce noise to the model and potentially reduce model accuracy.

1.6 Non-Unique SMILES Check

Intentions: Determine the distribution of SMILES that are non-unique and unique in the dataset. Non-unique SMILES appear more than once in the dataset.

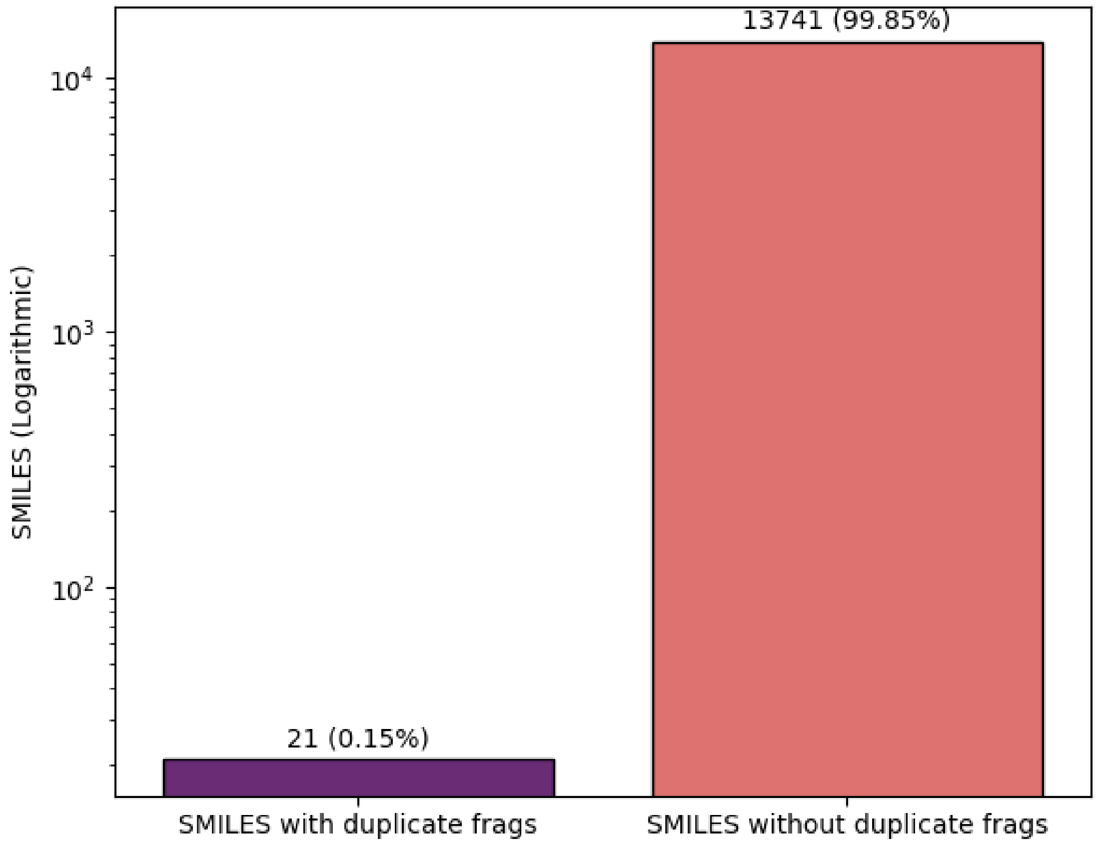


Findings: 3,127 SMILES were non-unique (22.7% of dataset) and 10,635 SMILes were unique (77.3% of dataset)

Interpretation: Remove the non-unique SMILES from the dataset to ensure that the data points are unique. This will prevent model-overfitting, as compared to not removing the non-unique SMILES, which will be redundant and memory-consuming.

1.7 Duplicate Fragment in SMILES Check

Intentions: Determine the distribution of SMILES with and without duplicate fragments in the dataset

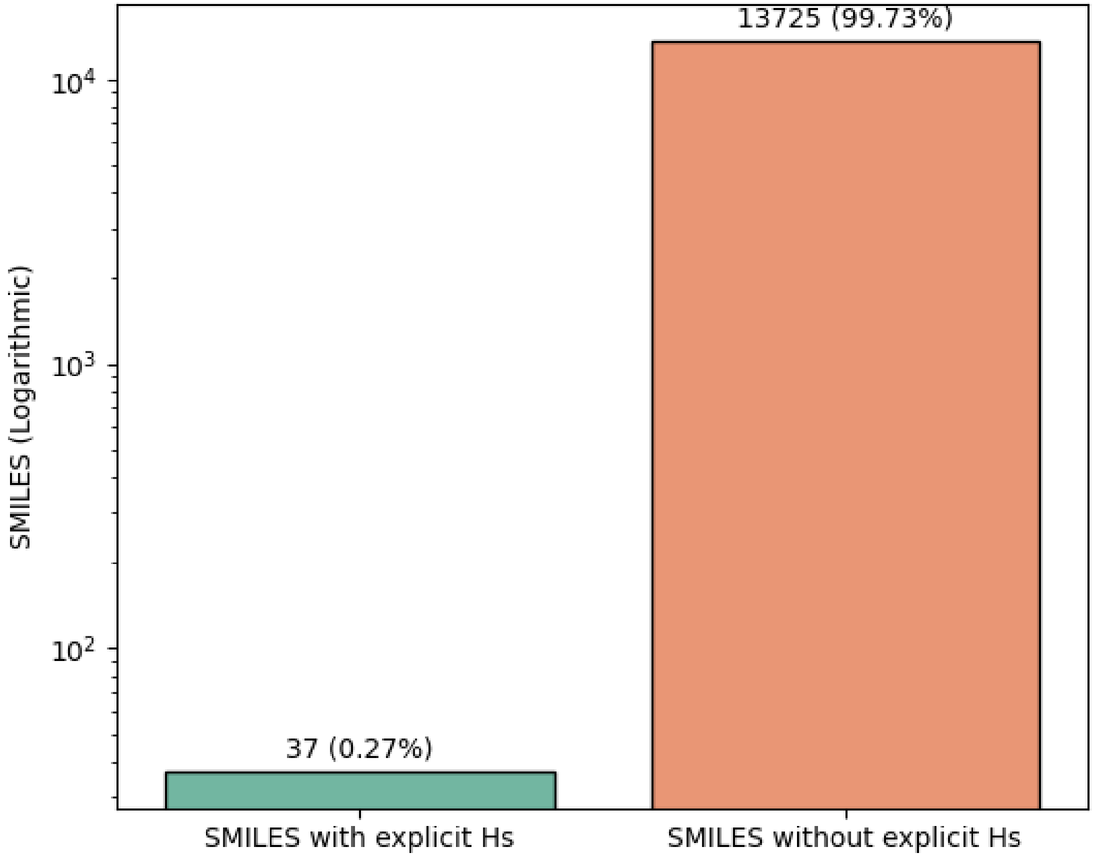


Findings: 21 SMILES had duplicate fragments (0.15% of dataset) and 13,741 SMILES had no duplicate fragments (99.85% of dataset)

Interpretation: Duplicate fragments increase the tokenization length, and therefore should be removed to decrease tokenization length and standardize with the dataset. In addition, the removal of duplicate fragments allows for the model to focus on the active moiety of the compound.

1.8 Explicit Hs in SMILES Check

Intentions: Determine the distribution of SMILES with explicit Hs and inexplicit Hs in the dataset. Explicit Hs are hydrogen atoms that are explicitly written in the SMILES of the compound.

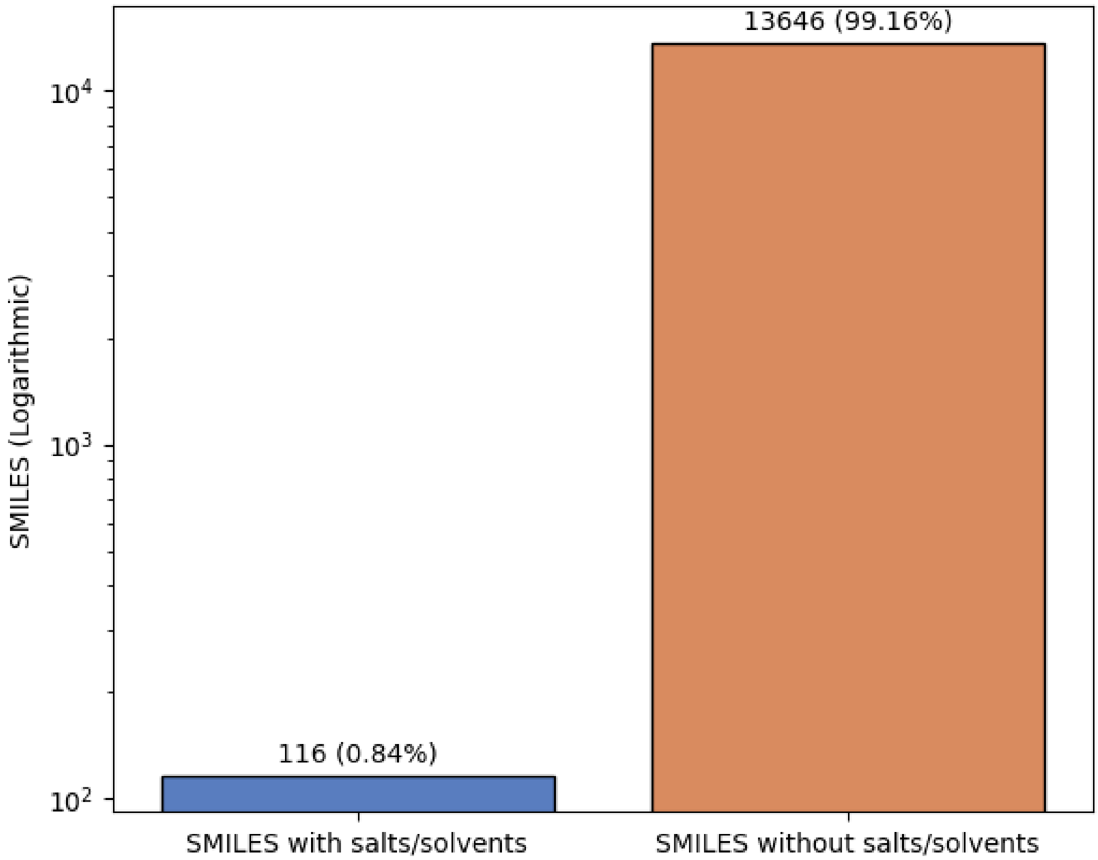


Findings: 37 SMILES had explicit Hs (0.27% of dataset) and 13,725 SMILES had inexplicit Hs (99.73% of dataset)

Interpretation: The SMILES with explicit Hs should be converted to inexplicit Hs for standardization with the dataset and a decrease in tokenization length. In addition, the removal of explicit Hs does not result in information loss as implicit Hs can be determined using the octet rule.

1.9 Salt/Solvents in SMILES Check

Intentions: Determine the distribution of SMILES with and without salts/solvents in the dataset.



Findings: 116 SMILES had salts/solvents (0.84% of dataset) and 13,646 SMILES didn't have salts/solvents (99.16 % of dataset)

Interpretation: Salts and solvents should be stripped from compounds because they are not a part of the active moiety, and only serve as external portions added for preparing compounds for ingestion. In addition, the stripping of salts and solvents would lead to shorter tokenization lengths, standardization with the dataset, and a focus on the active moiety.